Dr. Anika Sharma

Scientist E — Atmospheric Instrumentation & Applied AI

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PROFESSIONAL SUMMARY

Electronics and avionics engineer with twelve years of airborne-systems integration and qualification testing, followed by ten years leading scientific instrumentation and edge-AI development in government service. Experienced in MIL-STD-1553 test rigs, UAV payload electronics, sensor telemetry, embedded GPU deployment and technical procurement.

CORE COMPETENCIES

Avionics (MIL-STD-1553 / ARINC 429, Avionics integration) · Scientific instrumentation (Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems) · Embedded systems (Embedded C, ESP32 / STM32 microcontrollers) · Edge AI (NVIDIA Jetson deployment) · Drones & UAV systems (UAV payload integration) · AI / machine learning (PyTorch, Deep learning) · IoT (MQTT and Node-RED, InfluxDB and Grafana) · Computer vision (Image super-resolution, Object detection) · GPU computing (CUDA / GPU computing) · Model optimisation (TensorRT and ONNX, Pruning and knowledge distillation) · Software & tools (Python, Docker and Linux) · Technical leadership (Team leadership)

PROFESSIONAL EXPERIENCE

Scientist E, Upper Air Instrumentation Division, Kestrel Atmospheric Instrumentation Laboratory, New Delhi (07/2016 – Present)
Responsible for upper-air observation instruments, the division workshop and the laboratory's IoT and AI/ML activities.

- Lead a four-member AI/ML group developing lightweight super-resolution and detection models for UAV imagery, deployed on NVIDIA Jetson devices with TensorRT.
- Led acceptance testing and commissioning of radiosonde ground-receiving systems at three observing stations, writing the test protocol and the defect-tracking procedure.
- Designed and deployed an IoT monitoring network for laboratory infrastructure (ESP32 sensor nodes, MQTT, Node-RED, InfluxDB and Grafana dashboards) covering temperature, humidity and power quality.
- Prepared technical specifications and served on bid-evaluation committees for instrument procurement through the Government e-Marketplace in line with the General Financial Rules.
- Authored standard operating procedures for pre-flight checks and calibration of upper-air instruments used across the division.

Additional charge: **Workshop In-charge**, Upper Air Division Workshop — 04/2019 – Present
Additional charge of the division workshop supporting instrument repair, fabrication and calibration.

- Introduced a job-card and spares-inventory system for instrument repairs and calibration work.
- Supervise technicians and plan preventive maintenance for workshop test equipment.

Nodal officer: **Nodal Officer, IoT and AI/ML Initiatives** — 01/2021 – Present

Senior Design Engineer, Avionics Integration Group, Halcyon Avionics Design Bureau, Bengaluru (06/2004 – 06/2016)
Integration and qualification testing of airborne electronics and UAV payloads.

- Developed payload interface electronics for a fixed-wing UAV electro-optical sensor and supported its ground and flight trials.
- Integrated mission-computer and display units on MIL-STD-1553 avionics buses and built the automated test rig used for bench validation.
- Mentored graduate engineer trainees on avionics bus protocols and test documentation.

Graduate Engineer, Test Systems Group, Meridian Electronics Limited, Bengaluru (08/2002 – 05/2004)
Built automated test equipment for avionics line-replaceable units.

- Developed LabVIEW test sequences for power-supply and interface boards of avionics units.

KEY PROJECTS

Edge super-resolution pipeline for UAV imagery *(Ongoing)*, Principal investigator, Kestrel Atmospheric Instrumentation Laboratory (09/2021 – Present)
Research prototype that enhances low-resolution UAV video in real time on embedded GPUs using pruned and distilled deep networks.
Technologies: NVIDIA Jetson Xavier NX, NVIDIA Jetson Orin Nano, PyTorch, TensorRT, ONNX, Docker, Structured pruning, Knowledge distillation, INT8 quantisation

- Implemented structured channel pruning and knowledge distillation to shrink the super-resolution network for embedded deployment.
- Converted models through ONNX to INT8 TensorRT engines, achieving 3.1× faster inference on Jetson Xavier NX.

Radiosonde ground-station acceptance test protocol, Coordinator (01/2018 – 03/2019)
Station acceptance testing varied between sites.

- A common acceptance protocol and checklist adopted for new ground-receiving systems.

IoT monitoring of laboratory infrastructure, Project lead (02/2020 – 06/2022)
Instrument rooms lacked continuous environmental and power-quality monitoring.
Technologies: ESP32, Environmental sensors, MQTT, Node-RED, InfluxDB, Grafana

- Deployed ESP32 sensor nodes publishing over MQTT to a Node-RED flow, InfluxDB time-series storage and Grafana dashboards with threshold alerts.
- Continuous monitoring with automated alerts replaced manual twice-daily readings.

EDUCATION

Ph.D., Electrical and Electronics Engineering, Northfield Institute of Science and Technology (2021 – ongoing)
Thesis: Lightweight deep networks for real-time enhancement of aerial imagery

M.Tech., Avionics, Coastal Institute of Aeronautical Engineering, Chennai (2010)
Grade: CGPA 8.6/10

Fictional sample profile — not a real person

Thesis: Fault-tolerant data acquisition for airborne test instrumentation

B.E., Electronics and Communication Engineering, Northfield Institute of Science and Technology, Northfield University, Pune (2004)
Grade: First Class with Distinction

PUBLICATIONS

[1] **A. Sharma** and R. Iyer, "Knowledge-distilled lightweight networks for drone image super-resolution," accepted for publication in Example Transactions on Aerial Imaging.

[2] **A. Sharma**, K. Rao and R. Iyer, "Benchmarking embedded GPUs for real-time image enhancement," Example Symposium on Applied AI, pp. 88-93, Jan 2025.

[3] **A. Sharma** and R. Iyer, "Channel-pruned residual networks for real-time aerial image enhancement on embedded GPUs," Journal of Embedded Vision Systems, vol. 12, no. 3, pp. 45-58, Mar 2024. doi: 10.5555/jevs.2024.0123

[4] **A. Sharma**, P. Rao and S. Gill, "An IoT telemetry framework for upper-air ground stations," Proceedings of the Example Conference on Instrumentation (ECI 2023), pp. 112-117, Sep 2023.

[5] **A. Sharma** and S. Verma, "Edge telemetry dashboards for upper-air ground stations," Proceedings of the Example Conference on Instrumentation, Paper 17, Feb 2022.

[6] **A. Sharma**, P. Menon and R. Iyer, "A low-cost GNSS receiver chain for radiosonde tracking," Example Journal of Atmospheric Instrumentation, vol. 8, no. 2, pp. 101-112, Jun 2021. doi: 10.5555/ejai.2021.0042

MANUSCRIPTS UNDER REVIEW / IN PREPARATION

[1] **A. Sharma**, M. Khan and R. Iyer, "Detection-aware quality assessment of super-resolved drone images," under review at Example Transactions on Geoscience Imaging. *[Under review]*

[2] **A. Sharma**, "Simulation study of balloon-released gliding sondes," in preparation. *[In preparation]*

LEADERSHIP & RESPONSIBILITIES

Supervisor, M.Tech. interns in applied machine learning, Kestrel Atmospheric Instrumentation Laboratory (2022 – Present)
Supervise summer and dissertation interns on edge-AI projects.

Chair, Technical Evaluation Committee for radiosonde procurement, Kestrel Atmospheric Instrumentation Laboratory (04/2023 – 02/2024)
Chaired technical evaluation of bids for upper-air sounding systems.

- Evaluation report accepted by the competent authority.

AWARDS & HONOURS

Best Paper Award, Example Conference on Instrumentation (ECI 2023) (2023)

CERTIFICATIONS & TRAINING

Deep Learning for Edge Devices, Example Academy of Applied Computing (03/2022)
Duration: 40 hours
Credential: EAAC-DL-0042
Valid until: 03/2029

Workshop on Efficient Deep Learning for Embedded Vision, Example Conference on Instrumentation (ECI 2023) (12/2023)
Duration: 2 days

National Seminar on Upper-Air Observing Systems, Example Society of Atmospheric Instrumentation (08/2021)
Duration: 1 day

Mid-Career Training Programme for Scientists, Example Institute of Scientific Administration (sponsored by Ministry of Example Sciences) (02/2020)
Duration: 10 days

Public Procurement and GFR 2017, Example Institute of Public Administration (sponsored by Kestrel Atmospheric Instrumentation Laboratory) (11/2019)
Duration: 5 days

TECHNICAL SKILLS

Avionics: MIL-STD-1553 / ARINC 429, Avionics integration
Scientific instrumentation: Environmental qualification testing, Sensor calibration, Radiosonde and upper-air systems
Embedded systems: Embedded C, ESP32 / STM32 microcontrollers
Edge AI: NVIDIA Jetson deployment
Drones & UAV systems: UAV payload integration
AI / machine learning: PyTorch, Deep learning
IoT: MQTT and Node-RED, InfluxDB and Grafana

LANGUAGES

English, Hindi

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